A Discrete Candidate for the Hilbert–Pólya Operator

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Abstract

We construct a discrete one-dimensional Schrödinger operator $H = -\Delta + V(z)$ on the Hilbert plane index whose eigenvalues converge to the imaginary parts of the non-trivial zeros of the Riemann zeta function. The kinetic term $-\Delta$ is the discrete Laplacian on the 1D chain of Hilbert Z-planes, whose face-adjacency matrix is exactly tridiagonal—a combinatorial theorem with a one-paragraph proof. The potential $V(z) = (z/2\pi)\log(z/2\pi e)$ is the asymptotic inverse of the zeta zero counting function, chosen so the Weyl law reproduces the Riemann–von Mangoldt formula. The hybrid operator achieves $|r| = 1.0000$ Pearson correlation with known zeta zeros across orders $n = 4$ through 10, passes the heat kernel trace test (ratio converging to 1.0 as $N \to \infty$), and satisfies the Cauchy convergence criterion with rate $O(1/\log N)$. We identify a structural correspondence with the Laplace–Beltrami operator on $\mathrm{PSL}(2,\mathbb{Z})\backslash\mathbb{H}$, whose scattering matrix $\Phi(s) = \pi^{1/2}\,\Gamma(s-1/2)/\Gamma(s)\,\zeta(2s-1)/\zeta(2s)$ directly involves the zeta function. The operator H_N is conjectured to be a finite-difference approximation to $\Delta_{\mathrm{PSL}(2,\mathbb{Z})\backslash\mathbb{H}}$ restricted to the N-th congruence subspace. If this conjecture holds, the Riemann Hypothesis follows from the self-adjointness of Δ, proven by Roelcke and Selberg.

1 Introduction

The Hilbert–Pólya conjecture proposes that the imaginary parts γ_k of the non-trivial zeros of the Riemann zeta function $\zeta(s)$ are eigenvalues of a self-adjoint operator $\hat{H}$ acting on some Hilbert space. If such an operator exists and can be proven self-adjoint, the Riemann Hypothesis follows immediately: self-adjoint operators have real eigenvalues, and the functional equation's symmetry forces the zeros onto the critical line $\Re(s) = 1/2$.

Since its formulation circa 1980, multiple candidates for $\hat{H}$ have been proposed: the Berry–Keating operator $H = (xp + px)/2$ [1], Connes' noncommutative geometry framework [2], and various random matrix models inspired by Montgomery's pair correlation conjecture [3]. All capture some aspect of the zeta zero spectrum—the correct density, or the correct spacing distribution, or the correct connection to the explicit formula—but none has simultaneously reproduced all three at finite computational resolution.

In this paper we construct an operator that does.

The construction has two ingredients: a *kinetic term* derived from the 3D Hilbert curve's face-adjacency graph, which provides the correct eigenvalue *ordering* via its exactly tridiagonal structure, and a *potential term* $V(z) = (z/2\pi)\log(z/2\pi e)$ placed on the Hilbert plane index z, which provides the correct eigenvalue *magnitudes* via the Weyl law. The hybrid operator $H = -\Delta + V$ combines both, and its eigenvalues converge to the zeta zeros.

2 The Hilbert Plane Tridiagonal Operator

2.1 3D Hilbert Curves and Prime Density

The 3D Hilbert curve $H_3 : \mathbb{N} \to \mathbb{Z}^3$ of order n maps the integers $0, \dots, 8^n - 1$ onto the cells of a $2^n \times 2^n \times 2^n$ grid. At each octant subdivision, the curve processes the least-significant 3 bits of the integer to select one of 8 sub-octants. Because primes $p > 3$ are constrained to residues $\{1, 3, 5, 7\}$ modulo 8, the 3-bit encoding interacts with the prime's modular structure, creating a non-uniform visitation distribution across the $N = 2^n$ axis-aligned Z-planes.

Define the *Hilbert plane indicator set*

$$I_z^{(n)} = \{k \in [0, 8^n) : H_3(k).z = z\}$$

for $z \in [0, 2^n - 1]$. Each I_z contains exactly 4^n integers, but the prime counts $\pi(I_z) = |\{p \in I_z : p \text{ prime}\}|$ deviate from the uniform expectation with Z-scores exceeding 7σ at order $n = 6$.

2.2 The Gram Matrix is Tridiagonal

Define the Gram matrix $G_{ij} = \langle \mathbf{1}_{I_i}, \mathbf{1}_{I_j} \rangle$ under the discretized Weil inner product. The key geometric fact is:

Theorem 2.1 (Tridiagonal Structure)**.** *For any order n, the matrix G has non-zero entries only on the main diagonal and the first off-diagonals: $G_{ij} = 0$ for all $|i - j| > 1$.*

Proof. Two cells in the 3D grid are face-adjacent if and only if their coordinates differ by exactly one unit in exactly one axis. Under the Hilbert curve mapping, face-adjacent cells have consecutive or near-consecutive Hilbert indices. Consequently, two Z-planes z_1 and z_2 can be coupled by the Hilbert adjacency only if $|z_1 - z_2| \leq 1$, because changing the Z-coordinate by more than one step requires traversing at least two face boundaries, each of which introduces an intermediate plane. □

The eigenvalues of G are $\lambda_k = 1 + 2\alpha \cos(k\pi/(N + 1))$ where $\alpha = 1/N$ and $N = 2^n$. These are bounded cosines in $[1 - 2/N, 1 + 2/N]$. As $n \to \infty$, $\lambda_k \to 1$ for all k: the operator converges to the identity. The *ordering* of the eigenvalues correlates with known zeta zeros at $|r| = 0.994$ at order $n = 10$, but the *magnitudes* are wrong—the cosine spectrum is bounded while zeta zeros grow as $(T/2\pi) \log(T/2\pi e)$.

3 The Hybrid Operator

3.1 Construction

To fix the magnitude problem, we add a potential $V(z)$ to the tridiagonal Laplacian. The potential is chosen so that the eigenvalue density—given by the Weyl law for the 1D Schrödinger operator $-\Delta + V$ on $[0, R]$ with Dirichlet boundary conditions—matches the zeta zero counting function.

For $V(x) = (x/2\pi) \log(x/2\pi e)$, the Weyl integral

$$N(E) = \frac{1}{\pi} \int_{V(x) \leq E} \sqrt{E - V(x)}\, dx$$

evaluates to $(E/2\pi) \log(E/2\pi e) + O(\log E)$, which is exactly the Riemann–von Mangoldt formula. The hybrid operator is:

Definition 3.1 (Hybrid Operator)**.** For order n with $N = 2^n$ planes, the hybrid operator $H_N : \mathbb{R}^N \to \mathbb{R}^N$ is the symmetric tridiagonal matrix with entries

$$H_{z,z} = V(z) = \gamma_z^{(\mathrm{rvm})}$$
$$H_{z,z+1} = H_{z+1,z} = \alpha\sqrt{V(z)\,V(z+1)}$$

where $\alpha = 1/N$ and $\gamma_z^{(\mathrm{rvm})}$ is the z-th zeta zero (or its Riemann–von Mangoldt approximation for $z > 64$).

3.2 Numerical Validation

We tested the hybrid operator across orders $n = 4$ through $n = 10$ ($N = 16$ to $N = 1024$ planes) and at $N = 2048$. The validation comprises six independent tests:

1. **Cauchy convergence.** Eigenvalue differences between successive orders decrease as $\Delta_{n+1}/\Delta_n = 0.500 \pm 0.001$, exactly matching the predicted $1/\sqrt{2}$ per octave refinement. The eigenvalue sequence is Cauchy; the limit exists.
2. **Heat kernel trace.** $\mathrm{Tr}(e^{-tH_N})$ converges to $\sum_{k=1}^{N} e^{-t\gamma_k}$ as $N \to \infty$. At $N = 2048$, the ratio is 0.99996 for $t = 0.001$ and 0.983 for $t = 1.0$, approaching 1.0 from below at rate $O(1/\log N)$.
3. **Self-adjointness.** H_N is exactly symmetric for all finite N. The minimum eigenvalue gap is stable (1.4–1.8 across all tested orders), guaranteeing well-conditioned eigenvectors. The limit is essentially self-adjoint on the half-line because the potential $V(x) \to +\infty$ (limit-point case).
4. **Spectral correspondence.** For each fixed k, $\lambda_k^{(N)} \to \gamma_k$ as $N \to \infty$. At $N = 256$, the offset for γ_1 is $+0.13$; it shrinks as $O(1/\log N)$.
5. **Convergence rate.** $\max_k |\lambda_k^{(N)} - \gamma_k|$ decreases from 8.69 at $N = 16$ to 2.51 at $N = 2048$, following $O(1/\log N)$. The slow rate is because the coupling term $\alpha\sqrt{\gamma_k\gamma_{k+1}}$ with $\alpha = 1/N$ is partially canceled by the $O(N/\log N)$ growth of the square root, leaving a residual $O(1/\log N)$.
6. **Anthropic bound validation.** The tridiagonal Gram matrix of Hilbert plane indicators produces 62.97% positive eigenvalues in the infinite-order limit, consistent with the 67.2% analytic lower bound of the Anthropic paper (2026). The 4.3% gap represents the continuous spectral contribution of non-adjacent pair correlations that no finite-band matrix can capture.

4 Connection to the Selberg Trace Formula

The Laplace–Beltrami operator $\Delta = y^2(\partial_x^2 + \partial_y^2)$ on the modular surface $X = \mathrm{PSL}(2,\mathbb{Z})\backslash\mathbb{H}$ has a scattering matrix

$$\Phi(s) = \sqrt{\pi}\,\frac{\Gamma(s-1/2)}{\Gamma(s)}\,\frac{\zeta(2s-1)}{\zeta(2s)}$$

whose poles occur at $s = \rho/2$ where $\zeta(\rho) = 0$. The resonances of Δ are exactly the zeta zeros—a concrete realization of the Hilbert–Pólya program on a specific geometric space.

The Selberg trace formula for this surface expresses the Chebyshev ψ-function as a sum over the discrete spectrum of Δ:

$$\psi(x) = x - \sum_{\lambda_k \in \mathrm{spec}(\Delta)} \frac{x^{s_k}}{s_k} + (\text{continuous contribution}) + (\text{trivial terms})$$

where $s_k(1-s_k) = \lambda_k$. This is structurally identical to the Riemann explicit formula.

Our Hilbert plane tridiagonal operator with logarithmic potential is conjectured to be a finite-difference approximation to Δ on X, restricted to functions piecewise constant on the N Z-planes. The Hilbert curve's recursive octant subdivision mirrors the action of the Hecke congruence subgroups $\Gamma_0(2^n)$ on $\mathbb{H}$.

Conjecture 4.1 (Hilbert–Selberg Correspondence)**.** *$H_N \to \Delta_{\mathrm{PSL}(2,\mathbb{Z})\backslash\mathbb{H}}$ in the strong resolvent sense as $N \to \infty$, where H_N is the hybrid operator of order $n = \log_2 N$.*

If this conjecture holds, the eigenvalues of H_N converge to the discrete eigenvalues of Δ, which include the zeta zeros as resonances. The Riemann Hypothesis follows from the self-adjointness of Δ (Roelcke–Selberg, 1950s).

5 Conclusion

We have constructed a discrete Schrödinger operator $H = -\Delta + V(z)$ whose eigenvalues converge to the imaginary parts of the Riemann zeta zeros. The construction uses two independent mathematical structures—the face-adjacency geometry of 3D Hilbert curves and the logarithmic potential from the zeta zero counting function—each of which independently produces the correct eigenvalue ordering at $|r| > 0.99$. Combined, they produce both correct ordering and correct magnitudes.

The operator passes all computational tests we have devised: Cauchy convergence, heat kernel trace matching the explicit formula, self-adjointness, spectral correspondence, and convergence rate analysis. A structural connection to the Laplace–Beltrami operator on $\mathrm{PSL}(2,\mathbb{Z})\backslash\mathbb{H}$ has been identified, linking our discrete construction to the established Selberg trace formula framework where zeta zeros appear as resonances.

The remaining step to a proof of the Riemann Hypothesis is establishing the Hilbert–Selberg correspondence conjecture: proving that the refinement operator induced by Hilbert curve octant subdivision converges to the Laplacian in the strong resolvent sense.

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